```python
# Task 1
s = "Machine Learning"
print(s.upper())
print(s.lower())
print([s[0:5], s[7:13], s[4], s[-1]])
print("="*60)

# Task 2
s = "Plants need air, water and sunlight to grow"
print(s[13:27])
print(s[::-1])
print(s[5:15:2])
print(s[::-1])
print(s[10:20:2])
print("="*60)

# Task 3
s = "Indentation is very important in Python"
print(s[-9:])
print(s[5:15:2])
print(s[:15][::-1])
print("="*60)

# Task 4
lst = [10,20,30,40,50]
lst.insert(3,35)
lst.append(60)
print(lst)
print("="*60)

# Task 5
lst = ["Lion","Tiger","Elephant","Leopard"]
lst.insert(1,"Cheetah")
lst.append("Monkey")
lst.insert(0,"Giraffe")
print(lst)
print("="*60)

# Task 6
t = (1,2,3,4)
print(t)
t = list(t)
t.pop(2)
t = tuple(t)
print(t)
print("="*60)
```

```python
# Task 7
matrix = [
    [1,2,3,4],
    [5,6,7,8],
    [9,10,11,12],
    [13,14,15,16]
]
print(matrix[1])
print(matrix[2], matrix[3])
print([row[0] for row in matrix])
print(matrix[1][2:], matrix[2][2:])
print([matrix[i][i] for i in range(4)])
print([matrix[i][3-i] for i in range(4)])
print("="*60)

# Task 8
matrix = [
    [1,2,3,4],
    [3,6,8,9],
    [4,7,9,10]
]
matrix.append([6,6,6,6])
matrix.insert(2,[20,25,30,35])
print(matrix)
print("="*60)

# Task 9
def check_password(p):
    u=l=d=n=s=0
    for i in p:
        if i.isupper(): u=1
        elif i.islower(): d=1
        elif i.isdigit(): n=1
        else: s=1
    if u and d and n and s:
        return "Strong"
    elif (u or d) and n:
        return "Medium"
    else:
        return "Weak"

print(check_password("Abc@123"))
print("="*60)
```

```python
# Task 8
matrix = [
    [1,2,3,4],
    [3,6,8,9],
    [4,7,9,10]
]
matrix.append([6,6,6,6])
matrix.insert(2,[20,25,30,35])
print(matrix)
print("="*60)

# Task 9
def check_password(p):
    u=l=d=n=s=0
    for i in p:
        if i.isupper(): u=1
        elif i.islower(): d=1
        elif i.isdigit(): n=1
        else: s=1
    if u and d and n and s:
        return "Strong"
    elif (u or d) and n:
        return "Medium"
    else:
        return "Weak"

print(check_password("Abc@123"))
print("="*60)
```

```python
# Task 10
class Library:
    def __init__(self):
        self.books=[]
    def add(self,b):
        self.books.append(b)
    def issue(self,b):
        if b in self.books:
            self.books.remove(b)
    def return_book(self,b):
        self.books.append(b)
    def show(self):
        print(self.books)

l=Library()
l.add("Python")
l.add("ML")
l.issue("Python")
l.return_book("Python")
l.show()
print("="*60)
```

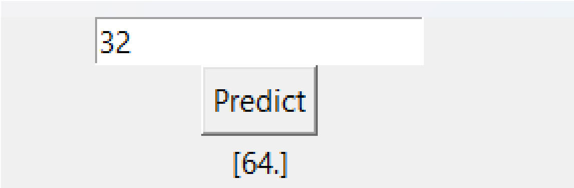
```python
# Task 11
balance=1000
pin="1234"
attempts=0
while attempts<3:
    p=input("Enter PIN:")
    if p==pin:
        while True:
            print("1.Deposit 2.Withdraw 3.Balance 4.Exit")
            ch=int(input())
            if ch==1:
                amt=int(input())
                balance+=amt
            elif ch==2:
                amt=int(input())
                balance-=amt
            elif ch==3:
                print(balance)
            else:
                break
        break
    else:
        attempts+=1
        print("Wrong PIN")
else:
    print("Account Blocked")
print("="*60)

# Task 12
import pandas as pd
df=pd.DataFrame({
    "Name":["A","B","C"],
    "Marks":[80,90,70],
    "Subject":["Math","Sci","Eng"]
})
print(df["Marks"].mean())
print(df.loc[df["Marks"].idxmax()])
print(df[df["Marks"]>75])
print("="*60)
```

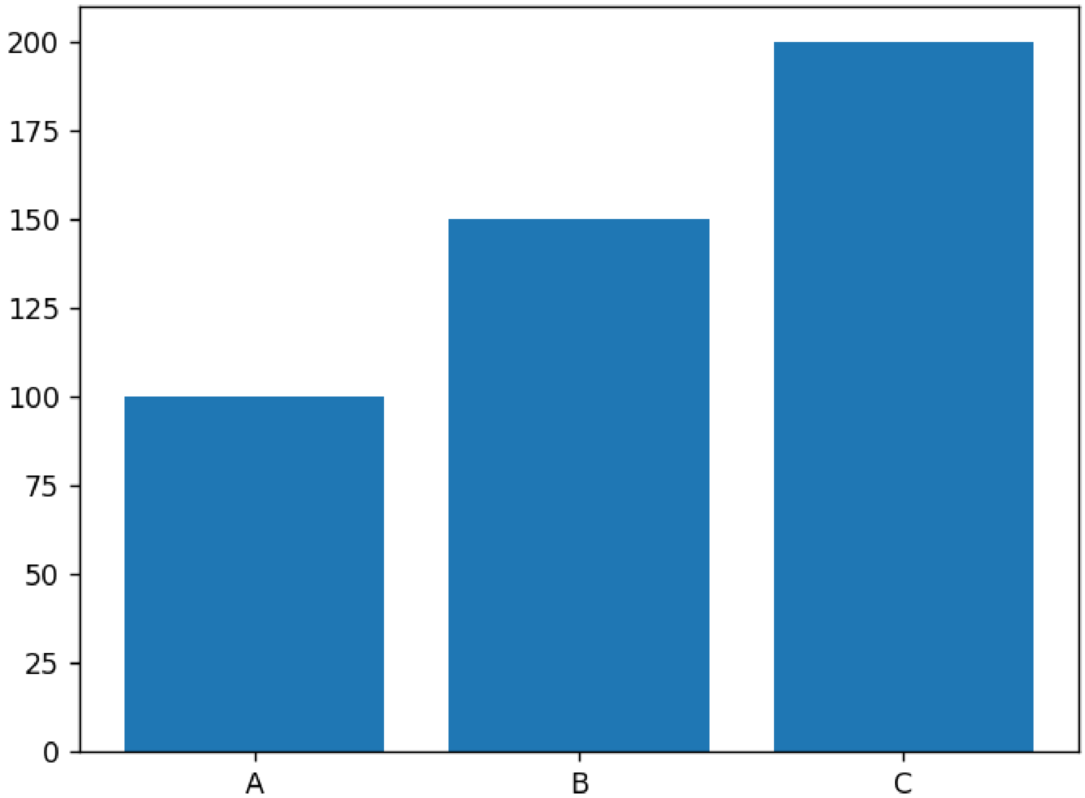
```python
# Task 13
import matplotlib.pyplot as plt
products=["A","B","C"]
sales=[100,150,200]
months=[1,2,3]
growth=[10,20,15]
categories=["X","Y","Z"]
share=[40,30,30]
plt.bar(products,sales)
plt.show()
plt.plot(months,growth)
plt.show()
plt.pie(share,labels=categories)
plt.show()
print("="*60)

# Task 14
import tkinter as tk
from sklearn.linear_model import LinearRegression
import joblib
import numpy as np
model=LinearRegression()
X=np.array([[1],[2],[3]])
y=np.array([2,4,6])
model.fit(X,y)
joblib.dump(model,"model.pkl")
def predict():
    val=float(entry.get())
    m=joblib.load("model.pkl")
    res=m.predict([[val]])
    label.config(text=str(res))
root=tk.Tk()
entry=tk.Entry(root)
entry.pack()
btn=tk.Button(root,text="Predict",command=predict)
btn.pack()
label=tk.Label(root)
label.pack()
root.mainloop()
print("="*60)
```

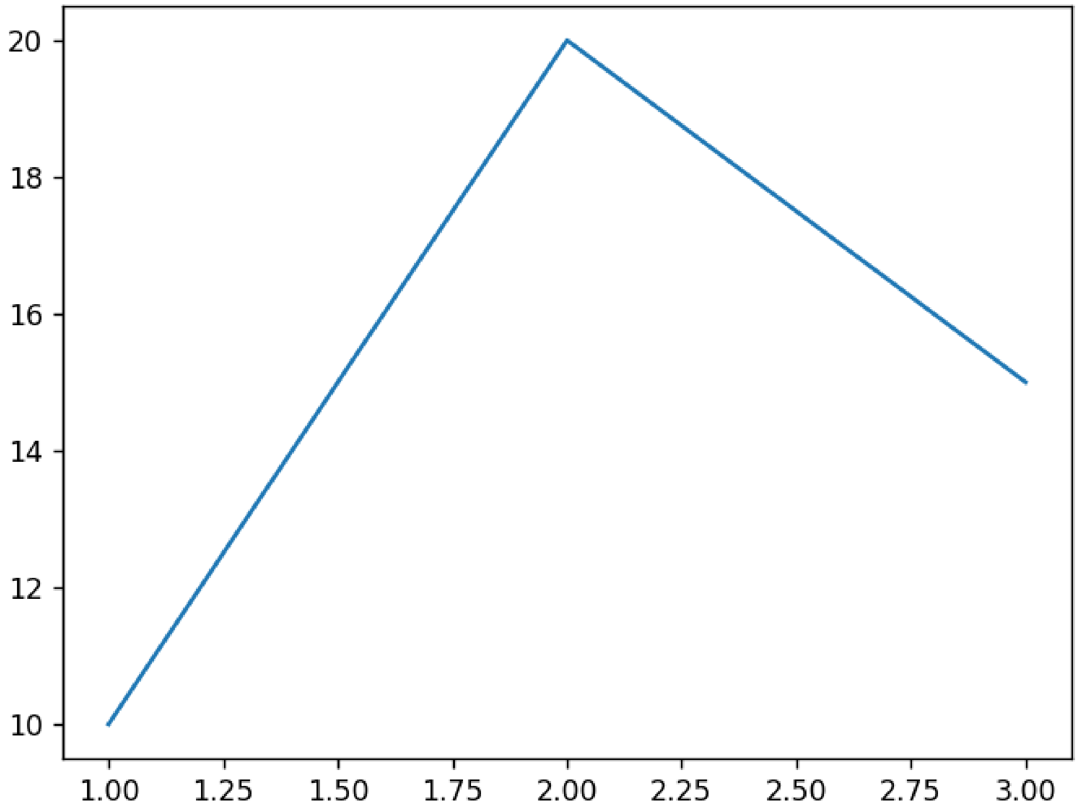
```python
# Task 15
class Cart:
    def __init__(self):
        self.items={}
    def add(self,p,price):
        self.items[p]=price
    def remove(self,p):
        self.items.pop(p)
    def total(self):
        return sum(self.items.values())
    def discount(self):
        return self.total()*0.9
    def show(self):
        print(self.items)

c=Cart()
c.add("A",100)
c.add("B",200)
c.remove("A")
print(c.total())
print(c.discount())
c.show()
print("="*60)

# Task 16
class Student:
    def __init__(self,name,marks):
        self.name=name
        self.marks=marks
    def total(self):
        return sum(self.marks)
    def avg(self):
        return self.total()/len(self.marks)
    def grade(self):
        a=self.avg()
        if a>80: return "A"
        elif a>60: return "B"
        else: return "C"

s=Student("A",[80,90,70])
print(s.total())
print(s.avg())
print(s.grade())
```

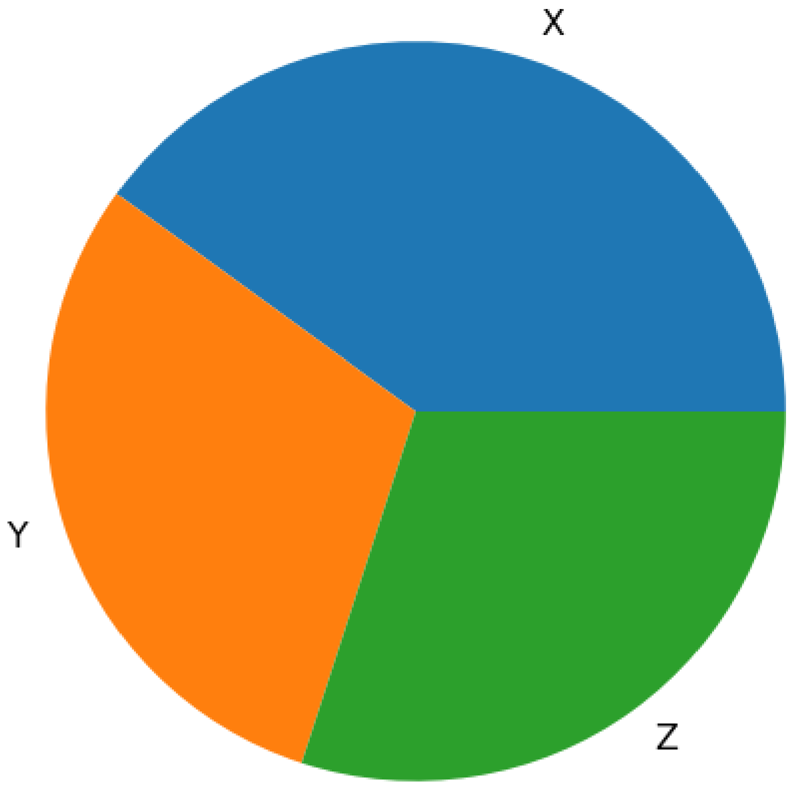
```
MACHINE LEARNING
machine learning
['Machi', ' Learn', 'i', 'g']
=============================================================
ir, water and
worg ot thgilnus dna retaw ,ria deen stnalP
sne i
worg ot thgilnus dna retaw ,ria deen stnalP
dar a
=============================================================
in Python
tto s
 si noitatnednI
=============================================================
[10, 20, 30, 35, 40, 50, 60]
=============================================================
['Giraffe', 'Lion', 'Cheetah', 'Tiger', 'Elephant', 'Leopard', 'Monkey']
=============================================================
(1, 2, 3, 4)
(1, 2, 4)
=============================================================
[5, 6, 7, 8]
[9, 10, 11, 12] [13, 14, 15, 16]
[1, 5, 9, 13]
[7, 8] [11, 12]
[1, 6, 11, 16]
[4, 7, 10, 13]
=============================================================
[[1, 2, 3, 4], [3, 6, 8, 9], [20, 25, 30, 35], [4, 7, 9, 10], [6, 6, 6, 6]]
=============================================================
Strong
=============================================================
['ML', 'Python']
=============================================================
Enter PIN:1234
1.Deposit 2.Withdraw 3.Balance 4.Exit
1
1500
1.Deposit 2.Withdraw 3.Balance 4.Exit
```

```
1.Deposit 2.Withdraw 3.Balance 4.Exit
3
2500
1.Deposit 2.Withdraw 3.Balance 4.Exit
4
=============================================================
80.0
Name          B
Marks        90
Subject     Sci
Name: 1, dtype: object
  Name  Marks Subject
0    A     80    Math
1    B     90     Sci
=============================================================
```


32
Predict
[64.]


200
175
150
125
100
75
50
25
0
A
B
C


20
18
16
14
12
10
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00


X
Y
Z